

# Wecamp Batch 11 - Capstone Presentation

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| Total Candidates | 43                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Format           | Divided into 6 teams (7+7+7+7+7+8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Judge Panel      | <a href="#">Liam Nguyen</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Capstone         | <div><h2>Overview</h2><p>Habit Tracker Pro is a web application that helps users build, track, and analyze their daily habits and personal goals.</p><p>The application covers:</p><ul style="list-style-type: none"><li>• Habit management</li><li>• Daily check-in tracking</li><li>• Goal setting with progress rules</li><li>• Streak and statistics dashboard</li><li>• Data persistence using mocked data or localStorage</li></ul><p>This capstone focuses on real-life business logic, state management, derived data, and user experience.</p><h2>Learning Objectives</h2><p>After completing this capstone, participants should be able to:</p><ul style="list-style-type: none"><li>• Model complex application state.</li><li>• Manage multiple related data domains.</li><li>• Implement derived data and calculations.</li><li>• Build clear, structured, and scalable React components.</li><li>• Explain trade-offs and design decisions.</li></ul><h2>Core Features (Mandatory)</h2><h3>1. Habit Management</h3><p>Users can manage all the habits they want to build.</p><p>Each habit includes:</p><ul style="list-style-type: none"><li>• Habit name</li><li>• Category:<ul style="list-style-type: none"><li>◦ Health</li><li>◦ Study</li><li>◦ Work</li><li>◦ Mindfulness</li><li>◦ Other</li></ul></li><li>• Frequency:<ul style="list-style-type: none"><li>◦ Daily</li><li>◦ Specific days of the week</li></ul></li><li>• Target per day (for example: drink water 8 times)</li><li>• Priority:<ul style="list-style-type: none"><li>◦ Low</li><li>◦ Medium</li><li>◦ High</li></ul></li><li>• Status:<ul style="list-style-type: none"><li>◦ Active</li><li>◦ Paused</li><li>◦ Archived</li></ul></li></ul><p>Users can:</p><ul style="list-style-type: none"><li>• Add / edit / delete habits</li><li>• Pause, resume, or archive a habit</li><li>• Filter habits by:<ul style="list-style-type: none"><li>◦ Category</li><li>◦ Frequency</li><li>◦ Priority</li><li>◦ Status</li></ul></li></ul></div> |

Habits with a missed check-in for the current day must be visually highlighted.

## 2. Daily Check-in Tracking

Users can record progress for each habit on each day.

Each check-in includes:

- Habit reference
- Date
- Completed count (for example: 5 of 8)
- Completion status:
  - Not Started
  - In Progress
  - Completed

Users can:

- Mark a habit as done for the day
- Increase or decrease the completed count
- Edit check-ins for the current day
- View check-ins grouped by date

The system must show daily completion progress in real time.

## 3. Goals & Progress Rules

Users can set a measurable goal for each habit.

Each goal includes:

- Target type:
  - Streak target (for example: 30 days in a row)
  - Total completions target (for example: 100 sessions)
- Target value
- Current progress (derived)

The application must enforce progress awareness.

For each goal:

- At 80% of the target: show an encouragement message
- At 100% of the target: show a goal-achieved alert

UI must clearly indicate goal status.

## 4. Streaks & Statistics Dashboard

A dedicated dashboard shows:

- Habits grouped by category
- Key indicators per habit:
  - Current streak
  - Longest streak
  - Total completions
  - Completion rate over the last 7 days
- Overall indicators:
  - % of habits completed today
  - Number of active habits
  - Number of habits at risk of breaking a streak

## 5. Data Persistence (Mocked Backend)

Use mocked JSON data or localStorage.

Data must persist after page refresh:

- Habits
- Check-ins
- Goals
- Progress and streak state

## Advanced Challenges (Mandatory)

## 6. Derived State & Performance

Avoid duplicated state.

All streaks, totals, percentages, and warnings must be derived from source data.

Clear separation between:

- Raw data (habits and check-ins)
- Computed values (streaks, completion rates, goal progress)

## 7. Undo / Reset Logic

Support at least one of the following:

- Undo last check-in action
- Reset all data to initial state

Logic must be clearly implemented and explained.

## 8. UX & Error Handling

Validate user input:

- Required fields cannot be empty
- Target value and counts cannot be negative
- Completed count cannot exceed the daily target
- A future date cannot be checked in

Show clear error messages.

UI must handle empty states gracefully:

- No habits
- No check-ins for the selected day
- No goals set

## Mock Data Example

```
{
  "habits": [
    {
      "id": 1,
      "name": "Drink water",
      "category": "Health",
      "frequency": "Daily",
      "targetPerDay": 8,
      "priority": "High",
      "status": "Active"
    }
  ],
  "checkins": [
    {
      "id": 1,
      "habitId": 1,
      "date": "2026-06-04",
      "completedCount": 5,
      "status": "In Progress"
    }
  ],
  "goals": [
    {
      "id": 1,
      "habitId": 1,
      "targetType": "Streak",
      "targetValue": 30
    }
  ]
}
```

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | <h2>Technical Constraints</h2> <ul style="list-style-type: none"> <li>• Frontend: React</li> <li>• No authentication</li> <li>• No real backend</li> <li>• State management must be clear and maintainable</li> <li>• Component structure must be logical and reusable</li> <li>• Data must persist using localStorage or mocked JSON</li> </ul> <h2>Bonus Features (Optional - Stretch)</h2> <ul style="list-style-type: none"> <li>• Calendar heatmap view of check-ins</li> <li>• Habit reminder notes per day</li> <li>• Export habit data to JSON file</li> <li>• Responsive mobile-first design</li> <li>• Read-only summary mode to share progress</li> <li>• Weekly progress chart by category</li> </ul> <h2>Deliverables</h2> <ul style="list-style-type: none"> <li>• GitHub repository</li> <li>• Working application</li> <li>• README including: <ul style="list-style-type: none"> <li>◦ Feature list</li> <li>◦ State structure explanation</li> <li>◦ Known limitations</li> </ul> </li> <li>• Demo, either live or recorded</li> </ul> |
| Agenda | <ul style="list-style-type: none"> <li>• Introduction</li> <li>• Presentation / Demo from Wecampers</li> <li>• Q&amp;A from teams, judges, and guests</li> <li>• Feedback from judges</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Results

### Team Presentation

| Criteria                 |         |         |         |         |         |         |         |         |         |         |         |         |
|--------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|                          | Judge 1 | Judge 2 | Judge 1 | Judge 2 | Judge 1 | Judge 2 | Judge 1 | Judge 2 | Judge 1 | Judge 2 | Judge 1 | Judge 2 |
| Technical (10)           |         |         |         |         |         |         |         |         |         |         |         |         |
| Teamwork (8)             |         |         |         |         |         |         |         |         |         |         |         |         |
| Decision making (8)      |         |         |         |         |         |         |         |         |         |         |         |         |
| Presentation + Demo (10) |         |         |         |         |         |         |         |         |         |         |         |         |
| Total score              |         |         |         |         |         |         |         |         |         |         |         |         |
| Average score            |         |         |         |         |         |         |         |         |         |         |         |         |

### Most Valuable Person